# HealthWise: Fuzzy Expert System for Coronary Heart Disease Diagnosis

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## Abstract

In this project, we designed a fuzzy expert system called **HealthWise** to estimate the risk of Coronary Heart Disease (CHD). The system uses fuzzy logic to model uncertainty in medical data. Three important medical indicators were used as input variables: blood pressure, cholesterol level, and heart rate.

Each input variable was represented using fuzzy membership functions such as low, normal, and high. The system uses a set of fuzzy rules to simulate medical reasoning and determine the CHD risk level. The Mamdani inference method was used to perform the fuzzy reasoning process.

The system was tested using three different patient scenarios provided by the instructor. The results show that fuzzy logic can effectively handle uncertain medical information and produce interpretable results for disease risk estimation.

## 1. Introduction

Coronary Heart Disease (CHD) is one of the most common cardiovascular diseases and a major cause of death worldwide. Early identification of patients who are at risk of CHD

is important for preventing serious health complications.

Traditional diagnostic systems often rely on fixed numerical thresholds. However, medical data is often uncertain or imprecise. For example, a blood pressure value may not always clearly indicate whether the patient is healthy or at risk.

Fuzzy logic provides a suitable approach for handling this type of uncertainty. Instead of using strict boundaries, fuzzy logic represents variables using linguistic terms such as **low**, **normal**, and **high**.

In this project, we built a fuzzy expert system that evaluates CHD risk using three physiological indicators:

- Blood Pressure
- Cholesterol Level
- Heart Rate

The system processes these inputs through fuzzy rules to determine whether the patient is **Healthy**, **At Risk**, or **Sick**.

## 2. System Overview

The system follows the standard fuzzy inference process, which consists of the following steps:

1. **Fuzzification**
2. **Rule Evaluation**
3. **Aggregation**
4. **Defuzzification**

The Mamdani fuzzy inference method was used because it is widely used in expert systems and

produces interpretable results.

The system was implemented using **Python** and the **scikit-fuzzy** library.

# 3. Input Variables

## 3.1 Blood Pressure

Blood pressure represents the force of blood against artery walls. High blood pressure is one of the main risk factors for heart disease.

The fuzzy categories used were:

- Low
- Medium
- High

The blood pressure values range approximately between **80 and 180**.

The triangular membership functions used for blood pressure are illustrated in Figure 1.

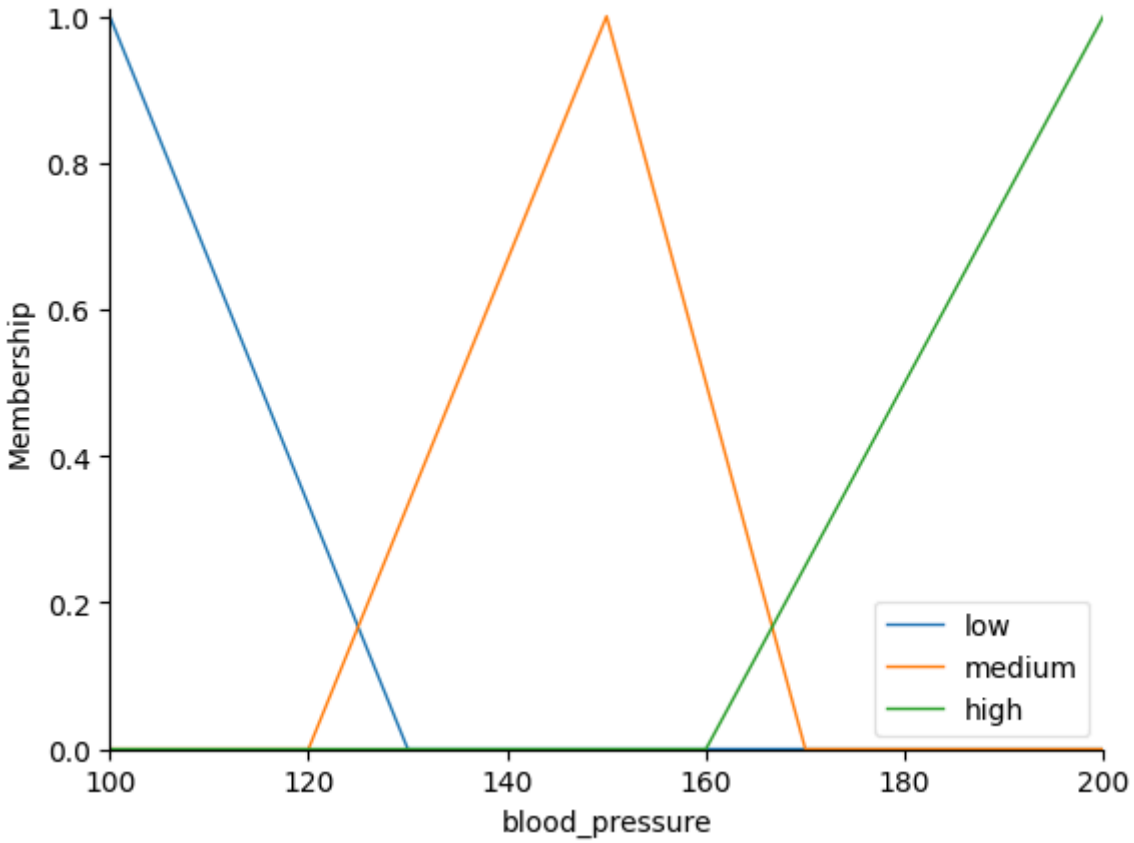


**Figure 1: Blood Pressure Membership Functions**

## 3.2 Cholesterol Level

Cholesterol level is another important indicator of heart disease risk. High cholesterol can lead to plaque buildup in the arteries.

The cholesterol values were divided into three fuzzy sets:

- Low
- High

The cholesterol range used in the system is approximately **100 to 300**.

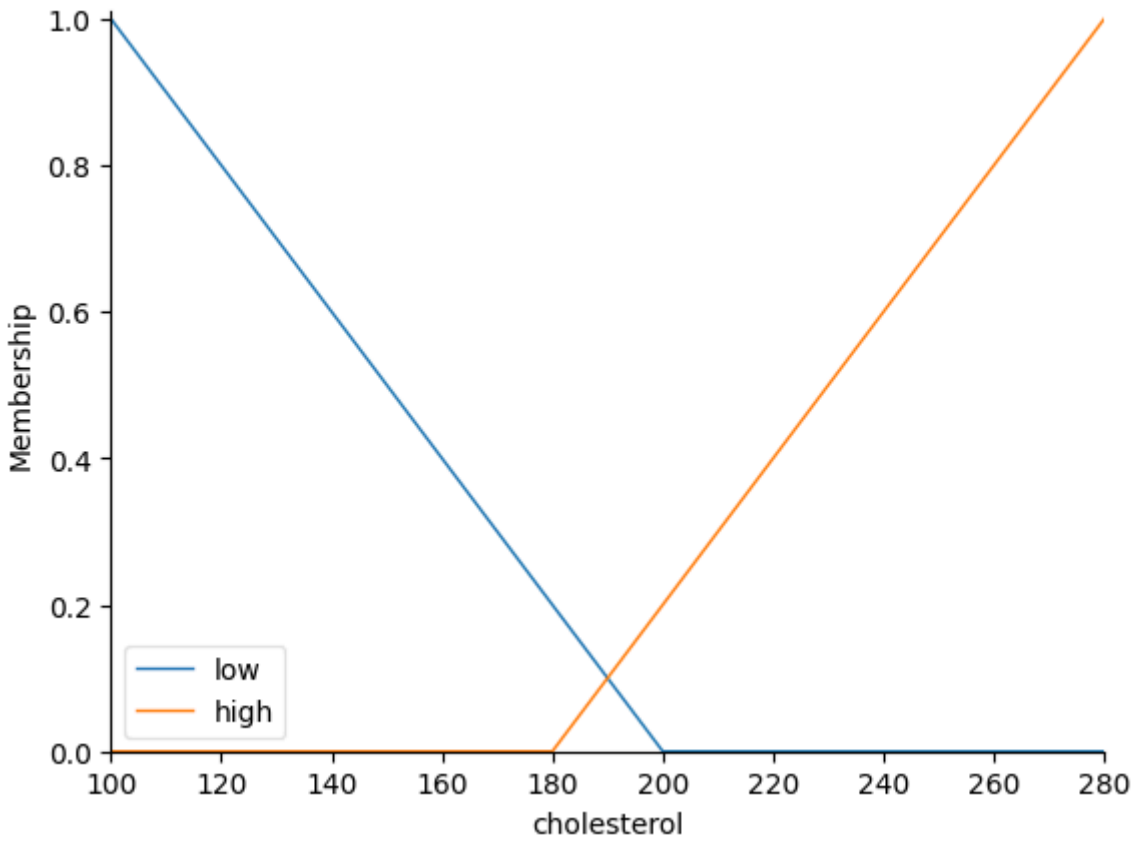


**Figure 2: Cholesterol Membership Functions**

## 3.3 Heart Rate

Heart rate measures the number of heart beats per minute. Abnormal heart rate values may indicate cardiovascular problems.

The heart rate was represented using three categories:

- Slow
- Moderate
- Fast

The range used for heart rate values is approximately 40 to 120 bpm.

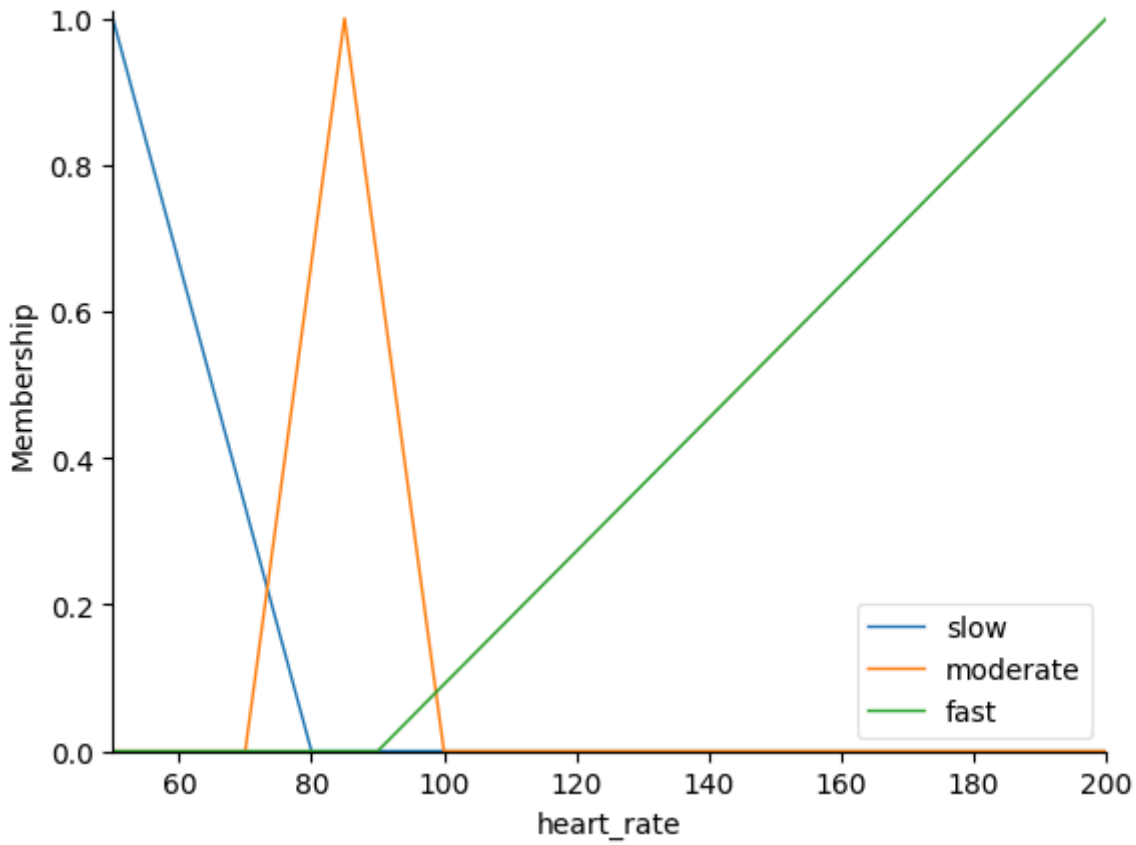


**Figure 3: Heart Rate Membership Functions**

# 4. Fuzzy Rule Base

The decision-making part of the system is based on a set of fuzzy rules that simulate expert medical reasoning.

The rules used in this system are:

**Rule 1**

If blood pressure is low and cholesterol is low and heart rate is slow

Then CHD risk is healthy

**Rule 2**

If blood pressure is low and cholesterol is low and heart rate is moderate

Then CHD risk is healthy

**Rule 3**

If blood pressure is medium and cholesterol is low and heart rate is moderate

Then CHD risk is middle

**Rule 4**

If blood pressure is medium and cholesterol is high and heart rate is slow

Then CHD risk is middle

**Rule 5**

If blood pressure is high and cholesterol is low and heart rate is moderate

Then CHD risk is sick

**Rule 6**

If blood pressure is high and cholesterol is high and heart rate is fast

Then CHD risk is sick

**Table 1: Fuzzy Rule Base**

| Rule | Blood Pressure | Cholesterol | Heart Rate | CHD |
|---|---|---|---|---|
| R1 | Low | Low | Slow | Healthy |
| R2 | Low | Low | Moderate | Healthy |
| R3 | Medium | Low | Moderate | Middle |

| | | | | |
|---|---|---|---|---|
| R4 | Medium | High | Slow | Middle |
| R5 | High | Low | Moderate | Sick |
| R6 | High | High | Fast | Sick |

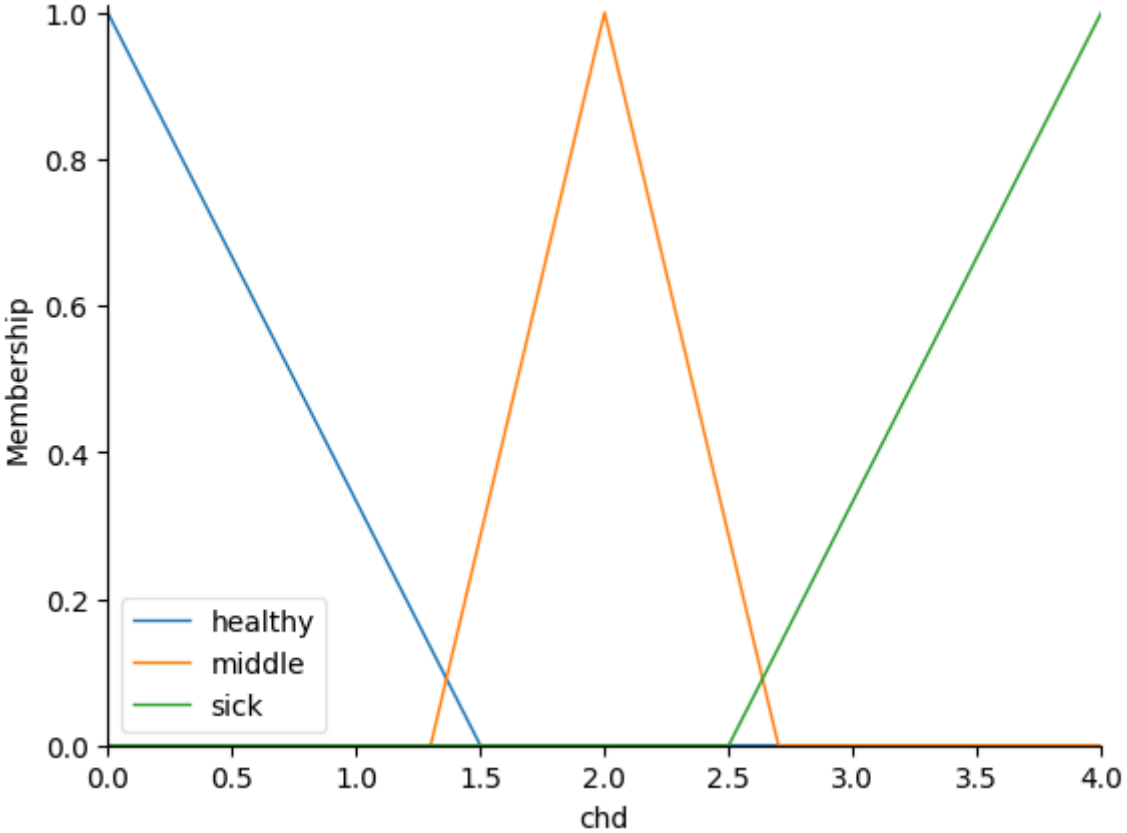


**Figure 4: CHD Risk Membership Functions**

# 5. System Implementation

The fuzzy system was implemented in Python using the **scikit-fuzzy library**.

The fuzzy expert system was implemented using the Python programming language and the scikit-fuzzy library. Python was chosen because it provides powerful scientific computing

libraries and allows easy implementation of fuzzy logic systems.

The implementation included the following steps:

1. Importing the required libraries
2. Defining fuzzy variables
3. Creating membership functions
4. Defining fuzzy rules
5. Building the control system
6. Running simulations for patient cases

# 6. Testing the System

| Patient | Blood Pressure | Cholesterol | Heart Rate | Result |
|---|---|---|---|---|
| Patient 1 | 105 | 160 | 55 | Healthy |
| Patient 2 | 120 | 195 | 65 | Healthy |
| Patient 3 | 165 | 186 | 95 | Middle Risk |

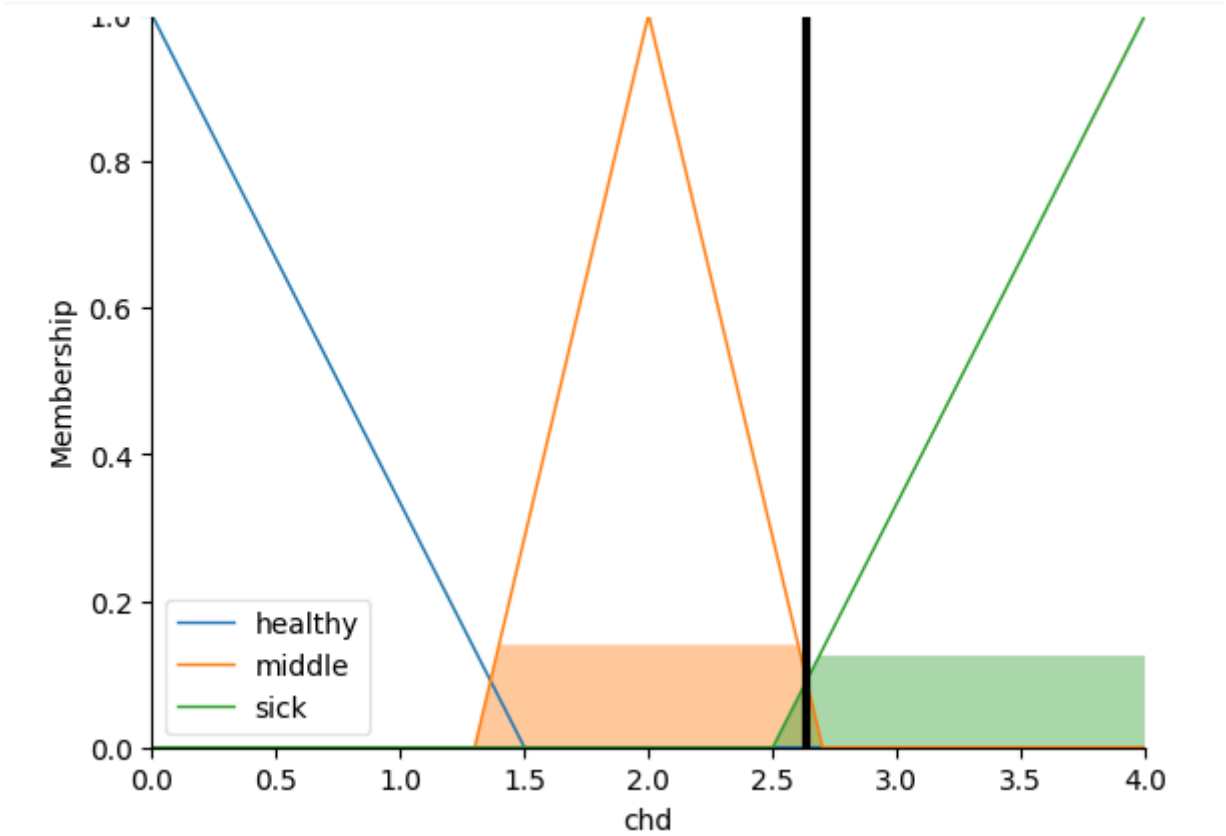


**Figure 5: Example of Fuzzy Inference Result**

**The fuzzy system produced the following crisp outputs:**

**Patient 1 → CHD = 0.612**
**Patient 2 → CHD = 0.731**
**Patient 3 → CHD = 2.637**

**Interpretation of CHD Values**
**0 – 1.5 → Healthy**
**1.3 – 2.7 → Middle Risk**
**2.5 – 4 → Sick**

# 7. Result Analysis

The results produced by the system show how fuzzy logic can evaluate risk levels based on different medical measurements.

For example:

- High blood pressure and high cholesterol values significantly increase the CHD

risk.

- Moderate values usually lead to an **at risk** classification.
- Low values in all parameters lead to a **healthy** result.

The fuzzy inference process begins with fuzzification of the input values. The system then evaluates the defined fuzzy rules using the minimum operator for the AND condition. The results of all activated rules are aggregated, and finally the centroid defuzzification method is applied to produce a crisp CHD risk value.

These results demonstrate that fuzzy logic can successfully model approximate reasoning similar to human medical judgment.

# 8. Mamdani vs Sugeno

Two common fuzzy inference methods are Mamdani and Sugeno.

The **Mamdani method** was chosen for this project because it provides outputs in the form of fuzzy sets and is easier to interpret.

The **Sugeno method** produces mathematical outputs and is often used in optimization or control systems.

For this medical risk estimation problem, Mamdani is more suitable because it provides clear linguistic interpretations such as healthy, at risk, and sick.

# 9. Conclusion

In this project, a fuzzy expert system was developed to estimate the risk of Coronary Heart Disease. The system uses blood pressure, cholesterol level, and heart rate as input variables.

Using fuzzy membership functions and a set of expert-defined rules, the system can classify patients into different risk levels.

The testing results demonstrate that fuzzy logic is an effective tool for modeling uncertainty in medical decision-making.

Such systems can assist medical professionals by providing decision support tools that help evaluate disease risk based on uncertain or approximate data.

# 10. Limitations

Although the system provides useful results, it has some limitations.

The system uses only three medical parameters. In real medical diagnosis, additional factors such as age, smoking habits, diabetes, and family history should also be considered.

Also, the number of rules in the system is limited.

Another limitation of the system is that the rule base does not cover all possible combinations of input values. As a result, some patient cases may not activate any rule in the system, which leads to missing predictions. Expanding the rule base would improve the robustness and reliability of the expert system.

# 11. Future Work

Future improvements to the system may include:

- Adding more medical parameters
- Increasing the number of fuzzy rules
- Integrating machine learning techniques

- Developing a graphical user interface for easier interaction

Future work may also include expanding the fuzzy rule base to cover more medical scenarios and ensure that all possible patient conditions can be evaluated.